

Rebuttal to Reviewer ngEh

We sincerely thank you for the thoughtful and constructive feedback. We are encouraged by your positive assessment of the core idea—particularly the formulation of symbolic regression via soft symbolic trees and differentiable variational inference—and we appreciate your openness to revising the score. Below, we address each of your concerns in detail.

Positioning w.r.t. LLM-based Symbolic Regression. We agree that LLM-based approaches represent an important and rapidly evolving direction in symbolic regression (SR). Our original submission focused primarily on probabilistic SR with principled uncertainty quantification, which led to emphasizing Bayesian and classical SR methods.

In the revision, we have: (i) added explicit discussion of LLM-based SR methods including LLM-SR (Shojaee et al., ICLR’25), DrSR (Wang et al., arXiv preprint 2025), LaSR (Grayeli et al., NeurIPS’24), and PhyE2E (Ying et al., Nature Machine Intelligence 2025) in the literature review; (ii) clarified the positioning of VaSST as complementary to various SR approaches including LLM-based architectures. However, VaSST focuses primarily on posterior inference over symbolic tree structures for learning scientific expressions directly from the data; and (iii) included a new Appendix section dedicated to the empirical comparison of VaSST with LLM-SR, using TinyLlama and Qwen as the underlying LLMs.

Key findings from empirical comparison of VaSST against LLM-SR. We conducted additional experiments with LLM-SR (Shojaee et al., ICLR’25), using two representative open-source language models: TinyLlama-1.1B-Chat and Qwen2.5-Coder-7B-Instruct. The task is to recover the Feynman equation

$$f = \beta(1 + \alpha \cos \theta),$$

with additive Gaussian noise having standard deviation 0.2. This particular Feynman equation has been newly added along with the equations considered in the original submission. Importantly, this setup reflects our central goal: learning the structure of scientific expressions directly from data, without assuming access to explicit domain-specific symbolic priors beyond minimal constraints.

We consider three prompting regimes of increasing informativeness, ranging from weak syntactic guidance to strongly structured scientific prompts. Across both models, we observe that LLM-SR is highly sensitive to prompt specification and initialization. Under weak or moderately informative prompts, both TinyLlama and Qwen frequently generate incomplete or non-executable functions (e.g., outputs containing only docstrings or missing functional bodies), leading to rejection during evaluation. Even under strong prompts, where the underlying scientific structure is described, the models occasionally produce expressions that partially capture the correct interaction (e.g., $\beta(\cdot + \alpha \cos \theta)$) but introduce extraneous transformations or violate execution constraints, resulting in degraded predictive performance relative to the noise level. These results suggest that LLM-based SR methods are powerful and complementary, but may require careful prompt design, model hosting, and filtering of invalid samples (scientific expressions). See the detailed prompts used and the results obtained from LLM-SR below.

In contrast, VaSST directly optimizes a differentiable probabilistic symbolic tree model. Since our goal is probabilistic SR with principled uncertainty quantification over symbolic structures,

VaSST provides a stable and computationally efficient framework that not only recovers interpretable expressions but also quantifies uncertainty at the level of symbolic forms, without relying on prompt engineering or large-scale generative sampling. The equation recovery results of VaSST while learning $f = \beta(1 + \alpha \cos \theta)$ across noiseless and Gaussian noise levels with $\sigma^2 \in (0.1^2, 0.2^2)$ shows exact recovery of the underlying symbolic structure with $K = 3, D = 3$ (maintaining similar experimental configurations as for the other Feynman equation in the original submission). The corresponding RMSEs also closely track the true noise levels added to the data. These additional experiments will be added to the revision. The results of VaSST for noise levels $\sigma \in (0.1, 0.2)$ are tabulated as follows.

VaSST Results for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ ($\sigma = 0.1$)			
We report the top 5 recovered symbolic expressions by VaSST for the noise level $\sigma = 0.1$.			
Rank	Trees	Regression coefficients	RMSE
1	$t_0 = \cos(2\alpha)$ $t_1 = \theta$ $t_2 = \beta + \beta\alpha \cos \theta$	$\beta_0 = -0.9293$	0.1651
		$\beta_1 = 0.0006$	
		$\beta_2 = 0.1066$	
		$\beta_3 = 1.0241$	
		$\beta_0 = -0.5573$	
2	$t_0 = \cos(\beta) + (\theta + \alpha) + 2\alpha$ $t_1 = \beta(\theta + \beta)$ $t_2 = \cos \theta(\beta + \beta\alpha)$	$\beta_1 = 0.0321$	0.5556
		$\beta_2 = 0.0506$	
		$\beta_3 = 0.8895$	
		$\beta_0 = -1.6734$	
		$\beta_1 = 0.0441$	
3	$t_0 = \cos(\beta + \theta) + (\beta + \theta)\beta\alpha$ $t_1 = (\beta\alpha + \cos \alpha) \cos \theta$ $t_2 = \beta \cos \alpha$	$\beta_2 = 1.0081$	0.7300
		$\beta_3 = -0.1732$	
		$\beta_0 = -2.9767$	
		$\beta_1 = 0.0477$	
		$\beta_2 = 0.0811$	
4	$t_0 = \theta + \theta\beta\theta$ $t_1 = \alpha$ $t_2 = \cos \theta \beta + \cos \theta \beta\alpha$	$\beta_3 = 0.8978$	0.7914
		$\beta_0 = -0.0451$	
		$\beta_1 = 0.1882$	
		$\beta_2 = -0.5556$	
		$\beta_3 = 1.0679$	
5	$t_0 = \beta + (2\beta + \alpha\theta)$ $t_1 = \cos \beta + 2\alpha$ $t_2 = \beta\alpha \cos \theta$		0.9053
Summary. The top-ranked model recovers the true structure through $t_2 = \beta + \beta\alpha \cos \theta$, with coefficient close to one.			

VaSST Results for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ ($\sigma = 0.2$)			
We report the top 5 recovered symbolic expressions by VaSST for the noise level $\sigma = 0.2$.			
Rank	Trees	Coefficients	RMSE
1	$t_0 = \cos(2\alpha)$ $t_1 = \theta$ $t_2 = \beta + \beta\alpha \cos \theta$	$\beta_0 = -0.9592$	0.2375
		$\beta_1 = -9.70 \times 10^{-5}$	
		$\beta_2 = 0.1103$	
		$\beta_3 = 1.0243$	
		$\beta_0 = -0.5533$	
2	$t_0 = \cos(\beta) + (\theta + \alpha) + 2\alpha$ $t_1 = \beta(\theta + \beta)$ $t_2 = \cos \theta(\beta + \beta\alpha)$	$\beta_1 = 0.0317$	0.5806
		$\beta_2 = 0.0506$	
		$\beta_3 = 0.8895$	
		$\beta_0 = -1.6789$	
		$\beta_1 = 0.0441$	
3	$t_0 = \cos(\beta + \theta) + (\beta + \theta)\beta\alpha$ $t_1 = (\beta\alpha + \cos \alpha) \cos \theta$ $t_2 = \beta \cos \alpha$	$\beta_2 = 1.0081$	0.7484
		$\beta_3 = -0.1743$	
		$\beta_0 = -2.9791$	
		$\beta_1 = 0.0478$	
		$\beta_2 = 0.0796$	
4	$t_0 = \theta + \theta\beta\theta$ $t_1 = \alpha$ $t_2 = \cos \theta \beta + \cos \theta \beta\alpha$	$\beta_3 = 0.8979$	0.8090
		$\beta_0 = -0.0454$	
		$\beta_1 = 0.1886$	
		$\beta_2 = -0.5575$	
		$\beta_3 = 1.0680$	
5	$t_0 = \beta + (2\beta + \alpha\theta)$ $t_1 = \cos \beta + 2\alpha$ $t_2 = (\beta \cos \theta)\alpha$		0.9220
Summary. The dominant contribution in the top-ranked model arises from $t_2 = \beta + \beta\alpha \cos \theta$ with coefficient close to one, while other trees have negligible influence.			

Prompts and Results of LLM-SR. We report the prompts used and the results obtained while learning the Feynman equation $f = \beta(1 + \alpha \cos \theta)$ (along with an additive Gaussian noise having standard deviation 0.2) using LLM-SR with TinyLlama-1.1B-Chat and Qwen2.5-Coder-7B-Instruct as the underlying LLMs.

LLM-SR Prompt Specifications with TinyLlama-1.1B-Chat for Feynman equation $f = \beta(1 + \alpha \cos \theta)$

We evaluate LLM-SR using TinyLlama-1.1B-Chat on the noisy Feynman equation

$$y = \beta(1 + \alpha \cos \theta) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.2^2).$$

For each prompt, the LLM is asked to evolve a NumPy function

`equation(beta, alpha, theta, params)`

and each candidate is scored by negative MSE after optimizing the free parameters using BFGS.

Prompt 1: Weak syntactic guidance.

- Objective: find a simple vectorized NumPy formula mapping (β, α, θ) to y .
- Allowed symbols: $\beta, \alpha, \theta, \text{params}[0:2], \text{np.sin}, \text{np.cos}$.
- Main constraints: exactly two lines, no new variables, no reassignment, correct output shape.
- Initialization:

$$y = c_0 \beta.$$

Prompt 2: Scientist-guided moderate prompt.

- Adds scientific context: angular distribution with possible forward-backward asymmetry.
- States that β acts as an overall scale, α may control angular asymmetry, and θ is angular.
- Encourages compact low-complexity expressions involving cosine dependence.
- Initialization:

$$y = c_0 \beta + c_1 \alpha.$$

Prompt 3: Strongly structured scientific prompt.

- States that the data follow an angular distribution with additive Gaussian noise of standard deviation 0.2.
- Provides the scientific structure

$$\text{response} = \text{scale} \times (\text{baseline} + \text{asymmetry} \times \text{angular factor}).$$

- Indicates that β is the scale, α controls asymmetry, and the angular factor is cosine-like.
- Initialization:

$$y = c_0 \beta (c_1 + c_2 \alpha \cos \theta).$$

LLM-SR Prompt Specifications with Qwen2.5-Coder-7B-Instruct for Feynman equation $f = \beta(1 + \alpha \cos \theta)$

We evaluate LLM-SR using Qwen2.5-Coder-7B-Instruct on the noisy Feynman equation

$$y = \beta(1 + \alpha \cos \theta) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.2^2).$$

Each candidate symbolic program is evaluated via negative MSE after parameter fitting using BFGS (reduced iterations for efficiency).

Prompt 1: Weak / fair prompt.

- Objective: learn a compact NumPy vectorized expression from data.
- Minimal guidance; only syntactic constraints imposed.
- Strict rules: two executable lines, no reassignment, valid shape, restricted operators.
- Initialization:

$$y = c_0 \beta.$$

Prompt 2: Scientist-guided prompt.

- Provides explicit scientific target:
- $$\beta(1 + \alpha \cos \theta).$$
- Explains roles of variables: β (scale), α (asymmetry), θ (angle).
 - Encourages recovery of interaction term $\beta \alpha \cos \theta$.
 - Emphasizes simple, interpretable structure and avoidance of noise overfitting.
 - Initialization:

$$y = c_0 \beta + c_1 \alpha.$$

Prompt 3: Strongly structured scientific prompt.

- Fully specifies the scientific structure and noise model.
- Encourages preserving multiplicative form:

$$\text{scale} \times (\text{baseline} + \text{angular modulation}).$$

- Strong inductive bias toward cosine-based interaction.
- Initialization:

$$y = c_0 \beta (c_1 + c_2 \alpha \cos \theta).$$

LLM-SR Results with TinyLlama-1.1B-Chat for Feynman equation $f = \beta(1 + \alpha \cos \theta)$

We evaluate TinyLlama-1.1B-Chat on

$$y = \beta(1 + \alpha \cos \theta) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.2^2).$$

Prompt	Evolved model from samples	Interpretation	Score/RMSE
Prompt 1	<code>def equation(...):</code>	Invalid / incomplete function.	Score = -10^{20}
Prompt 2	Only signature and docstring; no executable body. <code>def equation(...):</code>	Invalid / incomplete function.	Score = -10^{20}
Prompt 3	Only signature and docstring; no executable body. $\hat{y} = 1 + c_0\beta(c_1 + \alpha \cos \theta)$ generated via extra lines returning <code>sigma2 = 1 + y</code> .	Partially captures the correct angular structure, but violates the two-line constraint and adds an incorrect shift.	RMSE = 0.3748

Summary. TinyLlama is highly sensitive to prompting. Prompts 1–2 fail to produce executable symbolic bodies, while Prompt 3 recovers the relevant interaction form $\beta(\cdot + \alpha \cos \theta)$ but introduces extraneous transformations. The initialized Prompt 3 structure itself attains RMSE ≈ 0.2012 , close to the noise level, but the evolved TinyLlama output is less reliable.

LLM-SR Results with Qwen2.5-Coder-7B-Instruct for Feynman equation $f = \beta(1 + \alpha \cos \theta)$

We evaluate Qwen2.5-Coder-7B-Instruct on

$$y = \beta(1 + \alpha \cos \theta) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.2^2).$$

Prompt	Evolved model from samples	Interpretation	RMSE / Score
Prompt 1	<code>def equation(...):</code>	Invalid / incomplete function.	Score = -10^{20}
Prompt 2	Only signature and docstring generated. <code>def equation(...):</code>	Invalid / incomplete function.	Score = -10^{20}
Prompt 3	Only signature and docstring generated despite strong scientific guidance. <code>def equation(...):</code> Docstring with extended explanation and an additional “improved version” section, but no executable body.	Invalid / incomplete function.	Score = -10^{20}

Summary. Across all prompting regimes, Qwen frequently generates non-executable outputs consisting only of function signatures and descriptive text. Even with strong scientific prompting explicitly specifying the target structure, the model does not consistently produce valid symbolic expressions under the strict two-line constraint. However, the optimization logs indicate that the initialized scientific structure

$$\beta(1 + \alpha \cos \theta)$$

can achieve RMSE ≈ 0.2012 , close to the noise level, suggesting that the target form is representable but not reliably recovered through LLM-based generation.

Hyperparameter Transparency and Practical Guidance. We appreciate this suggestion. In the revision, we have: (i) added a dedicated summary table of hyperparameters in the Appendix, as well as a detailed description of each argument in the VaSST code in the README file of the anonymous GitHub repository (<https://anonymous.4open.science/r/VaSST-62C7>); and (ii) included practical heuristics, such as shallow depth with more trees for smooth functions, and deeper trees for complex compositional symbolic structure or improved predictive performance. A more principled heuristic for choosing (K, D) is given below, inspired by the ablation study.

Ablation Study for Number and Depth of Trees. We conducted an ablation study analyzing the impact of: (i) number of trees K and (ii) tree depth D .

For each of the Feynman equations considered, VaSST is implemented for different values of

$$K \in \{2, 3, 4, 5\}, \quad D \in \{2, 3, 4, 5\},$$

keeping all other hyperparameters fixed. These new results will feature in the Appendix, along with a discussion of VaSST’s sensitivity with respect to K and D .

A highlight of the ablation study for two representative Feynman equations

$$P = \frac{\kappa A(T_2 - T_1)}{d}, \quad \Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

is as follows. Gaussian noise with standard deviation 0.1 has been added to the data. See Tables 1—4. The key findings are:

- **Performance stabilizes with model complexity.** For both equations, RMSE improves significantly when moving from $K = 2$ to $K \geq 3$, and stabilizes for $K \geq 4, D \geq 3$. This indicates that VaSST does not require large model sizes to achieve optimal predictive accuracy.
- **Low-capacity regimes can limit symbolic recovery.** As expected, insufficient model capacity leads to occasional failures in recovering the true expression. For

$$P = \frac{\kappa A(T_2 - T_1)}{d},$$

failure occurs at $(K = 2, D = 5)$, and for

$$\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

failures occur at $(K = 2, D \in \{2, 4, 5\})$ and $(K = 3, D = 5)$. This highlights that both sufficient K and appropriate D are necessary for capturing compositional structure.

- **Runtime scales predictably with K and D .** Runtime increases approximately linearly in K and exponentially with D , reflecting the growth in the number of nodes. Importantly, moderate configurations (e.g., $K = 3, D = 3$) already achieve near-optimal RMSE, offering a favorable accuracy–efficiency trade-off.
- **Principled heuristics for choosing K and D .** Based on the ablation results, we include practical guidelines for selecting K and D . Since VaSST is designed to recover simple and interpretable scientific expressions, our recommendation follows an Occam’s razor principle: choose the smallest model capacity that attains stable RMSE and symbolic recovery. Empirically, $K \in \{3, 4\}$ and $D \in \{3, 4\}$ provide a favorable accuracy–interpretability–runtime trade-off, while larger depths increase runtime exponentially and may lead to unnecessarily complex symbolic expressions without substantial predictive gains.

Table 1: Runtime for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	50.6	97.2	191.3	401.6
3	62.1	112.0	221.8	498.5
4	72.2	125.0	249.8	628.2
5	83.1	146.0	299.4	803.4

Table 2: RMSE for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.1019	0.1002	0.1002	0.1070
3	0.1000	0.1011	0.1012	0.1019
4	0.1000	0.0999	0.1000	0.1000
5	0.0999	0.0999	0.0998	0.1007

Table 3: Runtime for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	54.7	98.8	192.4	399.6
3	61.5	112.0	221.4	491.3
4	67.9	123.1	251.8	600.9
5	72.3	134.8	286.4	800.5

Table 4: RMSE for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.2330	0.1012	0.1959	0.2349
3	0.1002	0.1019	0.1051	0.1918
4	0.1168	0.1002	0.1005	0.1012
5	0.1002	0.1005	0.1004	0.1010

Clarification on Discrete Symbolic Model Space Order. The number of splitting decision, feature, and operator combinations for each node of the symbolic tree is $2p|\mathcal{O}|$, while the total number of nodes is $N \cdot K$ where $N = 2^{D+1} - 1$. This yields a total complexity of

$$O((2p|\mathcal{O}|)^{N \cdot K}).$$

Dimensionality of Variational Parameter Space. In general, the total dimension of Θ is

$$|\mathcal{O}| + p + N \cdot K(1 + p + |\mathcal{O}|),$$

which accounts for $|\mathcal{O}|$ global operator weights in $\mathbf{w}_{\text{op}}$ and p global feature weights in $\mathbf{w}_{\text{ft}}$. In addition, for each of the K symbolic tree skeletons and each of its $N = 2^{D+1} - 1$ nodes, we have 1 parameter for the split decision, p parameters for the soft feature-selection weights, and $|\mathcal{O}|$ parameters for the soft operator-selection weights.

For instance, in the simulation experiment

$$y = 6 \sin(\mathbf{x}_0) \cos(\mathbf{x}_1)$$

provided in the notebook (<https://anonymous.4open.science/r/VaSST-62C7>), we have $p = 2$, $|\mathcal{O}| = 3$, $K = 3$, and $D = 3$. This configuration yields a total variational parameter dimension of Θ equal to 275.